

Tyler Johnston

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PROFESSIONAL SUMMARY

Software developer based in Lisbon with 2 years of professional experience building production workflow applications using C#/.NET, JavaScript/TypeScript, REST APIs, and Oracle SQL. MSc Information Management candidate at Nova IMS specializing in Business Intelligence, with experience across front-end, back-end, database systems, QA, production support, data-driven applications, and technical project delivery. *Portuguese work authorization via Portuguese student residency.*

EDUCATION

Nova IMS, Lisbon, Portugal

Sep 2025 - Feb 2027

Master's in Information Management | Business Intelligence Specialization — Currently Enrolled

Utah State University, Logan, Utah (USA)

Aug 2020 - May 2024

B.S. in Computer Science, Minor in Mathematics — Magna Cum Laude | GPA: 3.92 / 4.00 | Presidential Scholarship

SKILLS

Programming C#, JavaScript, TypeScript, Python, Java, C++, Oracle SQL, PostgreSQL

Frameworks & Tools .NET, REST APIs, React.js, Angular, Docker, Firebase, AWS, Power BI, Microsoft Fabric, Supabase

DevOps & QA Git, Agile/Scrum, CI/CD, debugging, application testing, QA, production support

Languages English (Native), Portuguese (B1)

EXPERIENCE

Full Stack Developer, Footprints, Inc. (aerospace/manufacturing software) | Logan, UT

Mar 2024 - Aug 2025

- Built and maintained production workflow applications for aerospace and steel-manufacturing engineering teams using C#/.NET, REST APIs, JavaScript, Ext JS, jQuery, Oracle SQL, 4D, and Visual Basic.
- Completed 240+ full-stack development tasks across Agile sprints, translating stakeholder requirements into front-end, back-end, database, API, and workflow automation features.
- Served as a dedicated QA owner for full-stack releases, testing front-end, back-end, API, and database changes before deployment and resolving defects through root-cause analysis.
- Optimized SQL queries, service-layer logic, and data retrieval workflows to improve reliability, maintainability, and performance across production systems.

Computer Vision Research Assistant, Utah State University | Logan, UT

Jan 2024 - May 2024

- Built computer vision pipelines using YOLOv8 and Roboflow to detect curb ramps, pedestrian signage, and traffic lights in Google Street View imagery for ADA compliance analysis.
- Automated image retrieval, GPS metadata extraction, and data preprocessing workflows to create geospatial datasets and heatmaps for city-scale accessibility evaluation.

Modern C++ Teaching Assistant, Utah State University | Logan, UT

Aug 2023 - Dec 2023

- Designed a custom C++ std::vector prototype using templates, iterators, and manual memory management to optimize large-dataset allocation and demonstrate advanced data structure design.
- Held weekly office hours and graded assignments for an upper-division C++ course, providing debugging support on memory management, templates, iterators, and STL containers.

Web Developer, Space Dynamics Laboratory | Remote

Mar 2023 - Jun 2023

- Developed a full-stack workflow design tool using React, React Flow, C#/.NET, and Docker microservices, enabling users to create process workflows through a responsive drag-and-drop interface instead of manual XML editing.
- Enabled JSON and XML workflow configuration import/export through Docker-integrated services, reducing manual configuration errors and improving developer productivity.

PROJECTS

Parallax Poker — Character-Driven Poker Roguelite

C# / Godot 4 / Python

- Built a C# / Godot 4 Texas Hold'em roguelite with a rules-correct poker engine covering all-ins, raise/reopening logic, uncalled chip refunds, O(1) hand evaluation, and asynchronous game-state management.
- Developed a personality-driven AI decision engine with 15+ behavior parameters, equity calculations, draw detection, board texture scoring, SPR logic, tilt states, and probabilistic betting decisions.
- Created a Monte Carlo balancing pipeline using headless Godot simulations, Python, Pandas, and Plotly to test 9 AI opponents across 288,000 simulated hands and identify win-rate outliers.

Flashcard Dungeon — Full-Stack Language Learning App (flashcarddungeon.com)

Angular / TypeScript / Supabase

- Built a full-stack Angular 17+ and TypeScript language-learning app for European Portuguese, combining spaced-repetition flashcards, RPG dungeon combat, achievements, shop upgrades, and persistent progress tracking.
- Designed a CEFR-mapped A1-B2 curriculum with directed node progression, prebuilt decks, pronunciation guidance from native Portuguese speakers, IndexedDB offline storage, and optional Supabase/PostgreSQL cloud sync.

Absentee Analysis — HR Absenteeism Dashboard

Python / Pandas

- Built a Vercel-hosted HR analytics tool that segmented employees into absenteeism risk groups using clustering models and a Random Forest classifier, enabling non-technical HR users to review risk segments and support data-driven workforce decisions.